114學年度大學部專題海報展

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Music Style Transfer to Chopin's Nocturne Style Using Genetic Algorithms

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Introduction

Chopin’s nocturnes are known for their lyrical melodies, expressive rubato, chromatic harmony, and emotionally rich phrasing, forming one of the most distinctive styles in Romantic piano literature. Genetic Algorithm (GA) is an evolutionary optimization method that iteratively refines candidate solutions through selection, crossover, and mutation, enabling exploration of complex, high-dimensional creative spaces such as symbolic music. This project applies a two-phase GA to perform music style transfer, transforming any input piano score into a Chopin-nocturne style. The system evolves melody and accompaniment separately, guided by a static Markov model trained on Chopin’s nocturnes, producing music that preserves the original structure while exhibiting nocturne-like expressivity and harmonic color.

Methodology

1. Representation: Each musical track is represented as a sequence of 16th-note time steps. Every step stores a symbolic value describing the musical state, where a number indicates the MIDI pitch, –1 denotes a rest or empty position, –3 represents the continuation of a sustained note, and NULL is used for alignment when no musical event is assigned to that position. Ornaments are encoded as integer offsets attached to the base pitch.

2. GA: A two-phase GA evolves the melody and accompaniment separately. The initial population is generated by applying light perturbations to the original score. Measure-aware crossover preserves rhythmic and harmonic structure, while mutation introduces controlled variation in pitch, rhythm, and ornamentation.

3. Fitness: The fitness function evaluates how closely the evolved music matches the nocturne style while remaining coherent. It integrates target-style similarity, musical syntax, and, in later generations, preservation of the original melody. Similarity is computed through feature distributions and structural patterns, while syntax reflects tonal consistency and harmonic smoothness. These combined criteria guide the GA toward producing nocturne-style results without losing the identity of the input piece.

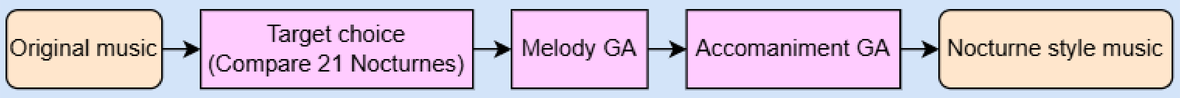


Figure 1: Workflow

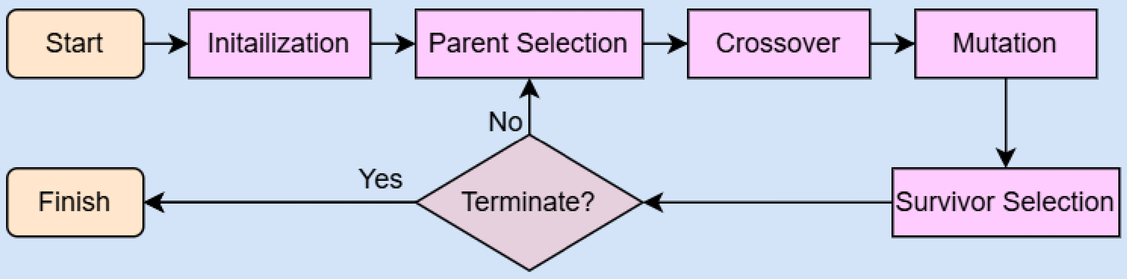


Figure 2: GA flowchart

Experiments

We conducted several experiments focusing on how different system configurations affect the generated results. These included trying different input–target combinations to examine how stylistic distance influences the GA’s behavior, comparing runs with and without the original-similarity term to understand its role in retaining the character of the input piece, modifying GA settings such as generation and population size, and adjusting the fitness weight distribution. Using Greensleeves as the final input, the system—through our similarity evaluation—automatically selected Chopin’s Nocturne No. 20 as the closest target, and the final configuration shown in Table 1 produced the smoothest and most coherent nocturne-like output.

Feature	Weight	Score
Target Similarity	0.45	0.1362
Structure Score	0.05	0.7545
Origin Similarity	1.0	0.6496
Syntax Score	1.5	0.9622

Table 1: The fitness weight setting



Figure 3: Greensleevess



Figure 4: Final result

Conclusion

The GA effectively captures key aspects of Chopin’s style, particularly in pitch and rhythm, and adapts them through crossover and mutation. However, note-level evolution limits the creation of entirely new structural sections, highlighting the complexity of higher-level musical form. Our experiments show that emphasizing musical consonance and syntax—or assigning greater weight to similarity with the original input—helps maintain expressive identity while achieving nocturne-style transformation. Overall, successful style transfer with GA requires balancing stochastic creativity with controlled evaluation to produce coherent, musically meaningful results.

Reference

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